

Supplemental Material for

Mutation-process priors calibrate protein language models for retrospective genotype–phenotype evolutionary ranking

S1 Codon-transition model

S1.1 Amino-acid substitution probability induced by nucleotide dynamics

We model a viral genome as a continuous-time Markov process (CTMP) on $\mathcal{B}^N$, where $\mathcal{B} = \{\text{A, C, G, U}\}$ is the RNA alphabet. The model assumes that sites evolve independently and that the transition rate at a site depends only on the current and mutated base. For a single base, let $Q = [q_{uv}]_{u,v \in \mathcal{B}}$ be the infinitesimal generator, with $q_{uu} = -\sum_{v \neq u} q_{uv}$. The transition probability over time T is

$$\mathbb{P}(b_{t+T} = v \mid b_t = u) = [\exp(TQ)]_{uv}. \quad (1)$$

For a codon $\mathbf{c} = (c_1, c_2, c_3)$, independence of the three nucleotide positions gives

$$\mathbb{P}(\mathbf{c}_{t+T} = \tilde{\mathbf{c}} \mid \mathbf{c}_t = \mathbf{c}) = \prod_{k=1}^3 [\exp(TQ)]_{c_k \tilde{c}_k}. \quad (2)$$

Let $g(\mathbf{c})$ denote the amino acid encoded by codon $\mathbf{c}$. The codon-induced probability of producing amino acid $\tilde{a}$ from wild-type codon $\mathbf{c}$ is therefore

$$p_T(\tilde{a} \mid \mathbf{c}) = \sum_{\tilde{\mathbf{c}}: g(\tilde{\mathbf{c}}) = \tilde{a}} \prod_{k=1}^3 [\exp(TQ)]_{c_k \tilde{c}_k}. \quad (3)$$

This is the CLIMB process prior used in the main text.

S1.2 Estimating the nucleotide transition matrix

For an ergodic CTMP, the long-run transition flux f_{uv} from base u to base v satisfies $f_{uv} = \pi_u q_{uv}$, where π_u is the stationary frequency [1]. We estimate

$$\hat{\pi}_u = \frac{n_u}{\sum_{u' \in \mathcal{B}} n_{u'}}, \quad \hat{f}_{uv} = \frac{n_{uv}}{\sum_{u' \neq v'} n_{u'v'}}, \quad \hat{q}_{uv} = \frac{\hat{f}_{uv}}{\hat{\pi}_u} \quad (u \neq v). \quad (4)$$

Rates are normalized so that $\sum_{u \neq v} \hat{f}_{uv} = 1$. Using the same stationary base frequencies as the Hie et al. sequence set [2],

$$\hat{\pi} = \begin{bmatrix} \text{A} & \text{C} & \text{G} & \text{U} \\ 0.2994 & 0.1837 & 0.1961 & 0.3208 \end{bmatrix}. \quad (5)$$

Combining these frequencies with substitution counts inferred from SARS-CoV-2 mutational data [3] gives

$$\hat{Q} = \begin{bmatrix} -0.3995 & 0.0480 & 0.2932 & 0.0583 \\ 0.1183 & -2.6735 & 0.0215 & 2.5338 \\ 0.3742 & 0.0693 & -1.3716 & 0.9281 \\ 0.0414 & 0.2934 & 0.0405 & -0.3753 \end{bmatrix}, \quad (6)$$

with row and column order (A, C, G, U).

S1.3 Context-independence check

The main model uses a context-averaged nucleotide process. To test whether this is a reasonable first-order approximation, we compared unconditional base-change probabilities with probabilities conditioned on the immediate 5' and 3' nucleotide contexts. The high C→U rate is stable across contexts, whereas A→G and G→A show stronger context variation. Thus, the context-averaged model captures the dominant central tendency but should be replaced by a context-dependent generator in future prospective versions.

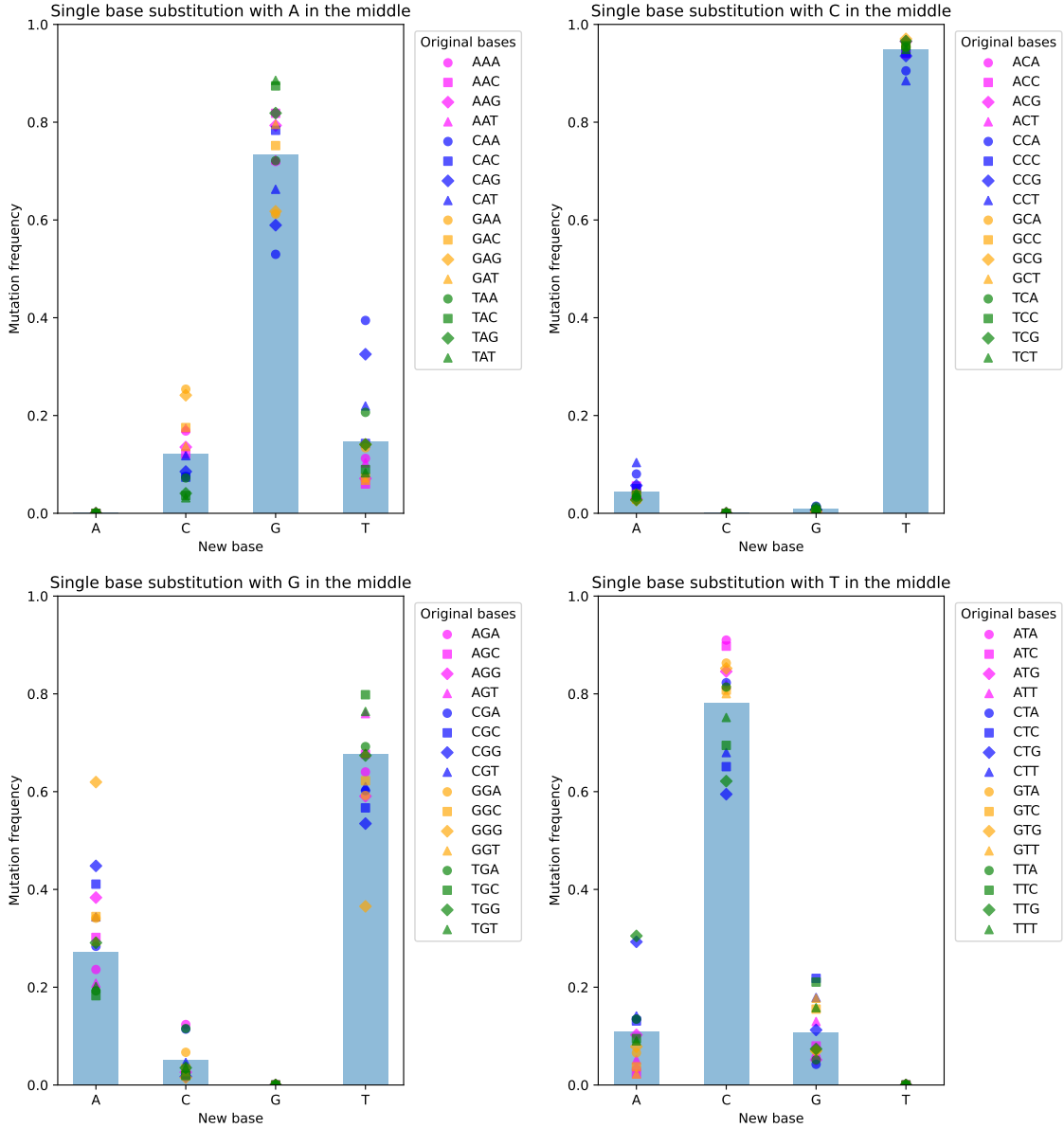


Figure S1: Unconditional and context-conditioned substitution probabilities. Bars show unconditional probabilities for each central-base substitution. Points show probabilities conditioned on the 16 possible trinucleotide contexts.

S2 CAC scoring and optimization

S2.1 Score definition

The main text defines CAC as a weighted mixture of standardized terms,

$$S_{\text{CAC}} = \alpha z(s_s) + \beta z(\log s_g) + \gamma z(\log p_T), \quad \alpha, \beta, \gamma \geq 0, \quad \alpha + \beta + \gamma = 1. \quad (7)$$

Here s_s is semantic change, s_g is grammaticality, and p_T is the codon-transition prior in Eq. (3). The standardization $z(\cdot)$ is performed over all nonsynonymous single-nucleotide substitutions of the reference Spike sequence.

S2.2 Validation sets

Weights were optimized by minimizing the escape mean rank (EMR) on a specified validation set. For a set $\mathcal{V}$ of known mutations,

$$\text{EMR}(\mathcal{V}; \alpha, \beta, \gamma) = \frac{1}{|\mathcal{V}|} \sum_{m \in \mathcal{V}} \text{rank}(m \mid \alpha, \beta, \gamma). \quad (8)$$

The current results are retrospective benchmarks. A strict prospective deployment should use time-split or leave-one-variant-out tuning to separate weight selection from final testing.

Table S1: Validation sets used for retrospective optimization and evaluation.

Name	Point mutations
DMS	K417E, K444Q, V445A, N450D, Y453F, L455F, E484K, G485D, F486V, F490L, F490S, Q493K, H655Y, R682Q, R685S, V687G, G769E, Q779K, V1128A
Alpha	N501Y, A570D, D614G, P681H, T716I, S982A, D1118H
Beta	L18F, D80A, D215G, K417N, E484K, N501Y, D614G, A701V
Delta	T19R, G142D, L452R, T478K, D614G, P681R, D950N
Omicron	A67V, T95I, G142D, L212I, G339D, S371L, S373P, S375F, K417N, N440K, G446S, S477N, T478K, E484A, Q493R, G496S, Q498R, N501Y, Y505H, T547K, D614G, H655Y, N679K, P681H, N764K, D796Y, N856K, Q954H, N969K, L981F
Combined	A570D, A67V, A701V, D1118H, D215G, D614G, D796Y, D80A, D950N, E484A, E484K, F486V, F490L, F490S, G142D, G339D, G446S, G485D, G496S, G769E, H655Y, K417E, K417N, K444Q, L18F, L212I, L452R, L455F, L981F, N440K, N450D, N501Y, N679K, N764K, N856K, N969K, P681H, P681R, Q493K, Q493R, Q498R, Q779K, Q954H, R682Q, R685S, S371L, S373P, S375F, S477N, S982A, T19R, T478K, T547K, T716I, T95I, V1128A, V445A, V687G, Y453F, Y505H

S2.3 Simplex search for optimal weights

The rank transform makes EMR piecewise constant and non-differentiable. We therefore used recursive subdivision of the simplex $\Delta^2 = \{(\alpha, \beta, \gamma) : \alpha, \beta, \gamma \geq 0, \alpha + \beta + \gamma = 1\}$. At each iteration the current triangle is divided into four subtriangles (Fig. S2). For each subtriangle T with vertices v_1, v_2, v_3 and centroid c , we evaluate

$$S(T) = \frac{1}{12} [\text{EMR}(v_1) + \text{EMR}(v_2) + \text{EMR}(v_3) + 9 \text{EMR}(c)], \quad (9)$$

and continue with the subtriangle having the smallest $S(T)$.

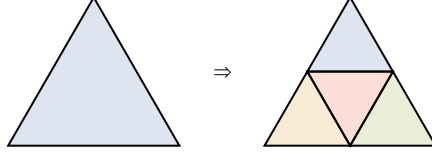


Figure S2: Recursive subdivision of the coefficient simplex.

S3 Model backbones and computational environment

We evaluated four PLM backbones: the viral Hie-BiLSTM model [2] and ESM-2-150M, ESM-2-650M, and ESM-2-3B [4]. For Hie-BiLSTM, we used precomputed semantic and grammaticality scores from the original authors. ESM-2 scores were generated on a server with a single NVIDIA H100 80GB GPU, two AMD EPYC 9124 16-core CPUs, and 768GB RAM, running Ubuntu 24.04.2 LTS. The Python environment used Python 3.10.17, PyTorch 2.7.0, Transformers 4.51.3, and CUDA 12.8.

S4 Additional performance results

Table S2: Comprehensive mean-rank comparison of CSCS and CAC.

Validation set	Model backbone	Escape rank (mean $\pm$ s.d.)		Wilcoxon p -value
		CSCS	CAC	
DMS	Hie-BiLSTM	3757.16 $\pm$ 2573.64	3493.47 $\pm$ 2973.51	0.55
DMS	ESM-2-150M	9387.26 $\pm$ 6127.43	5098.11 $\pm$ 3667.28	6.2×10^{-3}
DMS	ESM-2-650M	8562.32 $\pm$ 6869.83	5143.42 $\pm$ 3701.19	1.3×10^{-2}
DMS	ESM-2-3B	9130.00 $\pm$ 6500.32	5037.16 $\pm$ 3586.46	3.1×10^{-3}
Alpha	Hie-BiLSTM	7032.43 $\pm$ 5084.98	3567.86 $\pm$ 3281.72	0.11
Alpha	ESM-2-150M	10427.86 $\pm$ 6253.14	6732.86 $\pm$ 4672.34	0.19
Alpha	ESM-2-650M	12948.43 $\pm$ 5753.63	6816.29 $\pm$ 4391.33	0.11
Alpha	ESM-2-3B	10766.14 $\pm$ 5597.26	6480.71 $\pm$ 4743.41	0.11
Beta	Hie-BiLSTM	4046.00 $\pm$ 2348.84	1339.75 $\pm$ 1893.99	2.7×10^{-2}
Beta	ESM-2-150M	7902.00 $\pm$ 5731.57	3545.25 $\pm$ 3618.94	7.8×10^{-3}
Beta	ESM-2-650M	7932.25 $\pm$ 4493.72	3368.38 $\pm$ 3302.71	2.0×10^{-2}
Beta	ESM-2-3B	5013.25 $\pm$ 2767.09	2697.88 $\pm$ 2765.35	2.7×10^{-2}
Delta	Hie-BiLSTM	8102.29 $\pm$ 7945.03	5107.57 $\pm$ 7242.10	0.15
Delta	ESM-2-150M	11119.86 $\pm$ 5348.50	7255.43 $\pm$ 4137.39	7.8×10^{-3}
Delta	ESM-2-650M	11190.29 $\pm$ 5828.41	7094.71 $\pm$ 3428.05	3.9×10^{-2}
Delta	ESM-2-3B	10644.57 $\pm$ 4260.09	6154.71 $\pm$ 2643.14	2.3×10^{-2}
Omicron	Hie-BiLSTM	7655.30 $\pm$ 5208.90	3258.77 $\pm$ 3557.71	7.0×10^{-6}
Omicron	ESM-2-150M	10366.23 $\pm$ 5749.98	4261.67 $\pm$ 3283.74	7.0×10^{-6}
Omicron	ESM-2-650M	8687.47 $\pm$ 5740.13	4140.77 $\pm$ 3198.91	1.5×10^{-5}
Omicron	ESM-2-3B	8680.50 $\pm$ 6125.22	4059.87 $\pm$ 3336.31	8.5×10^{-5}
Combined	Hie-BiLSTM	6259.70 $\pm$ 5377.95	3841.07 $\pm$ 4183.94	2.4×10^{-4}
Combined	ESM-2-150M	9815.72 $\pm$ 6052.50	5308.55 $\pm$ 4280.29	1.3×10^{-6}
Combined	ESM-2-650M	8857.37 $\pm$ 6263.02	5326.83 $\pm$ 4195.95	1.6×10^{-5}
Combined	ESM-2-3B	8735.83 $\pm$ 6094.85	5204.23 $\pm$ 4054.37	8.6×10^{-6}

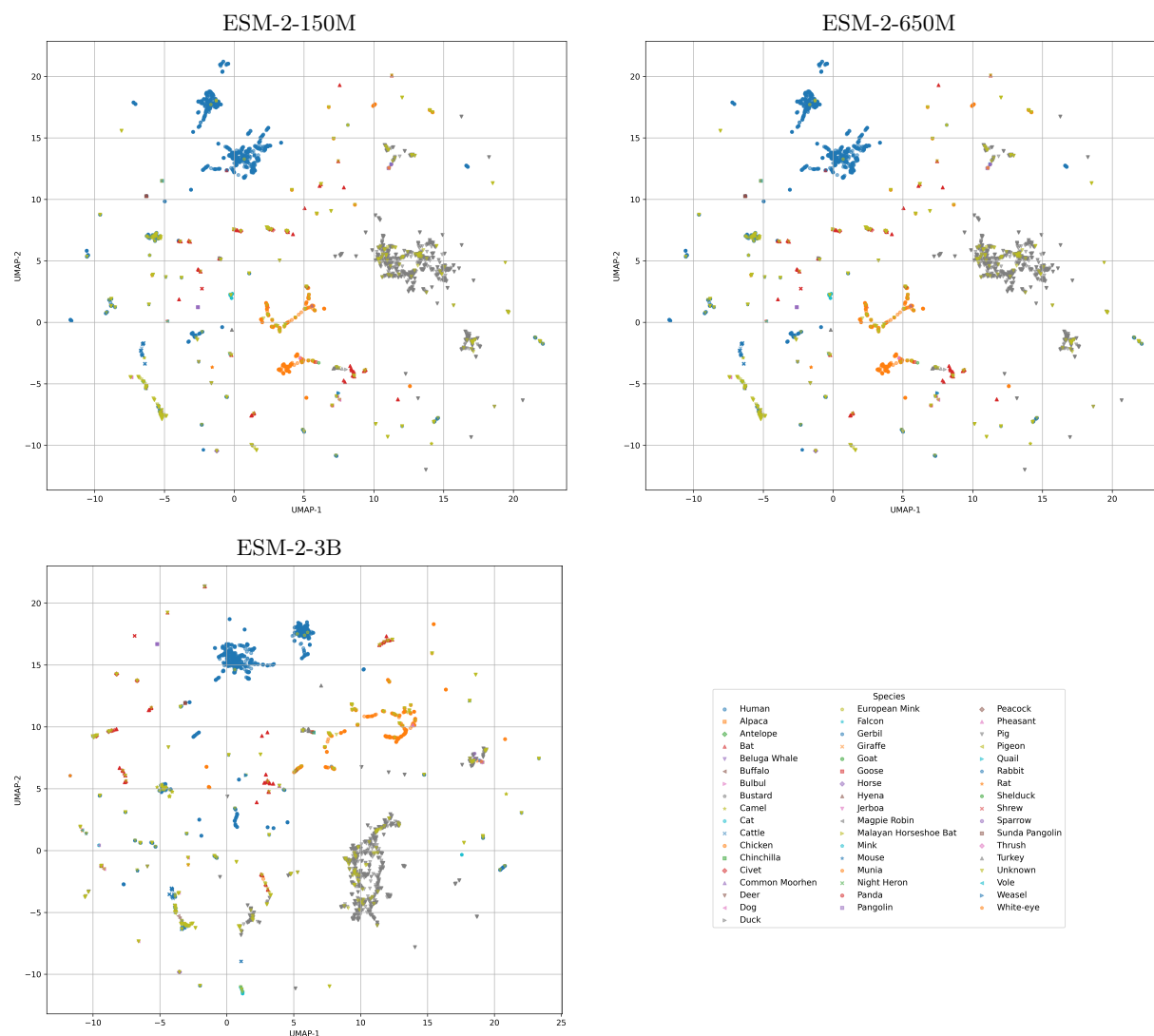


Figure S3: UMAP projections of ESM-2 Spike embeddings. Points represent unique SARS-CoV-2 Spike sequences colored by host species.

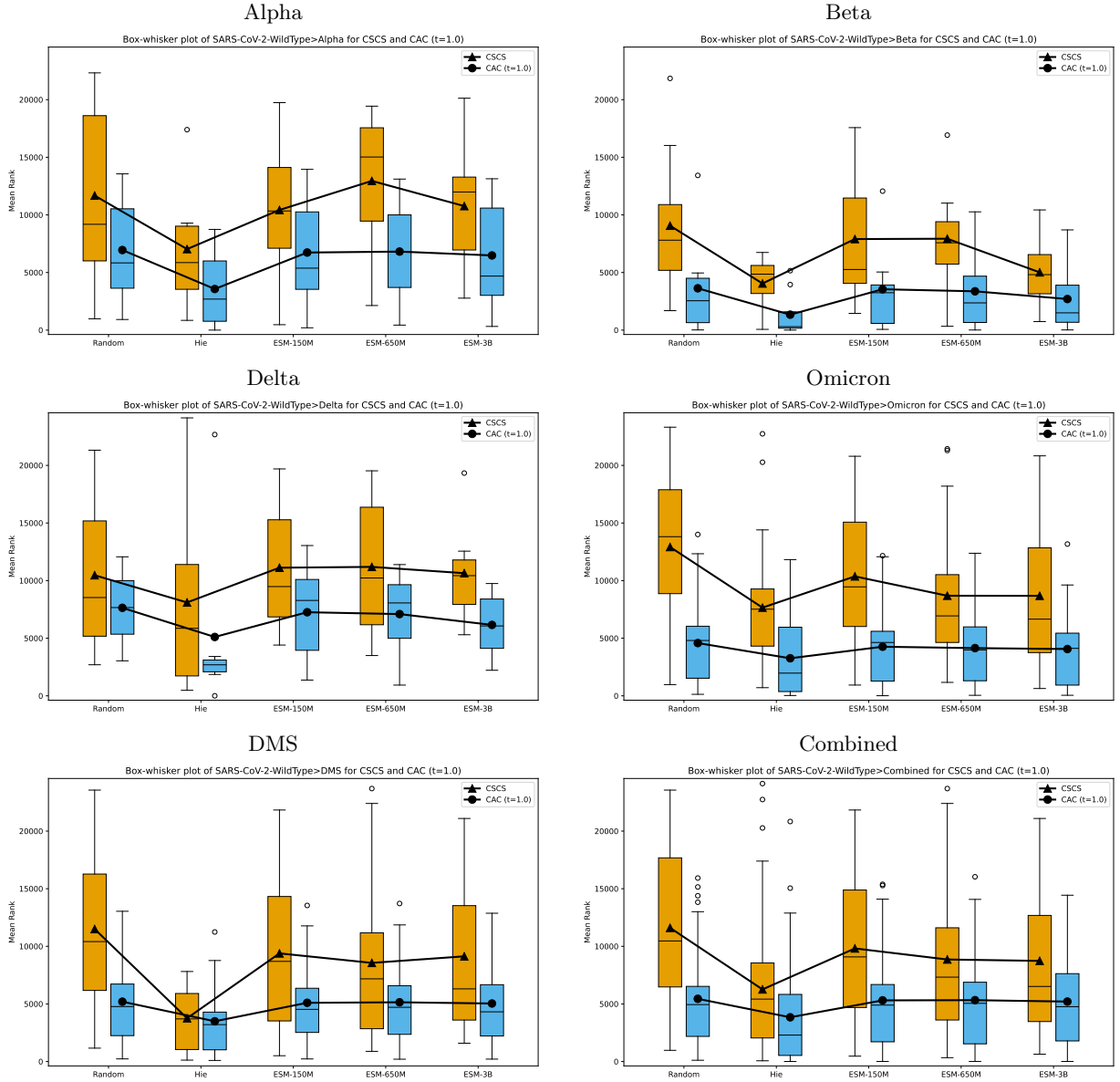
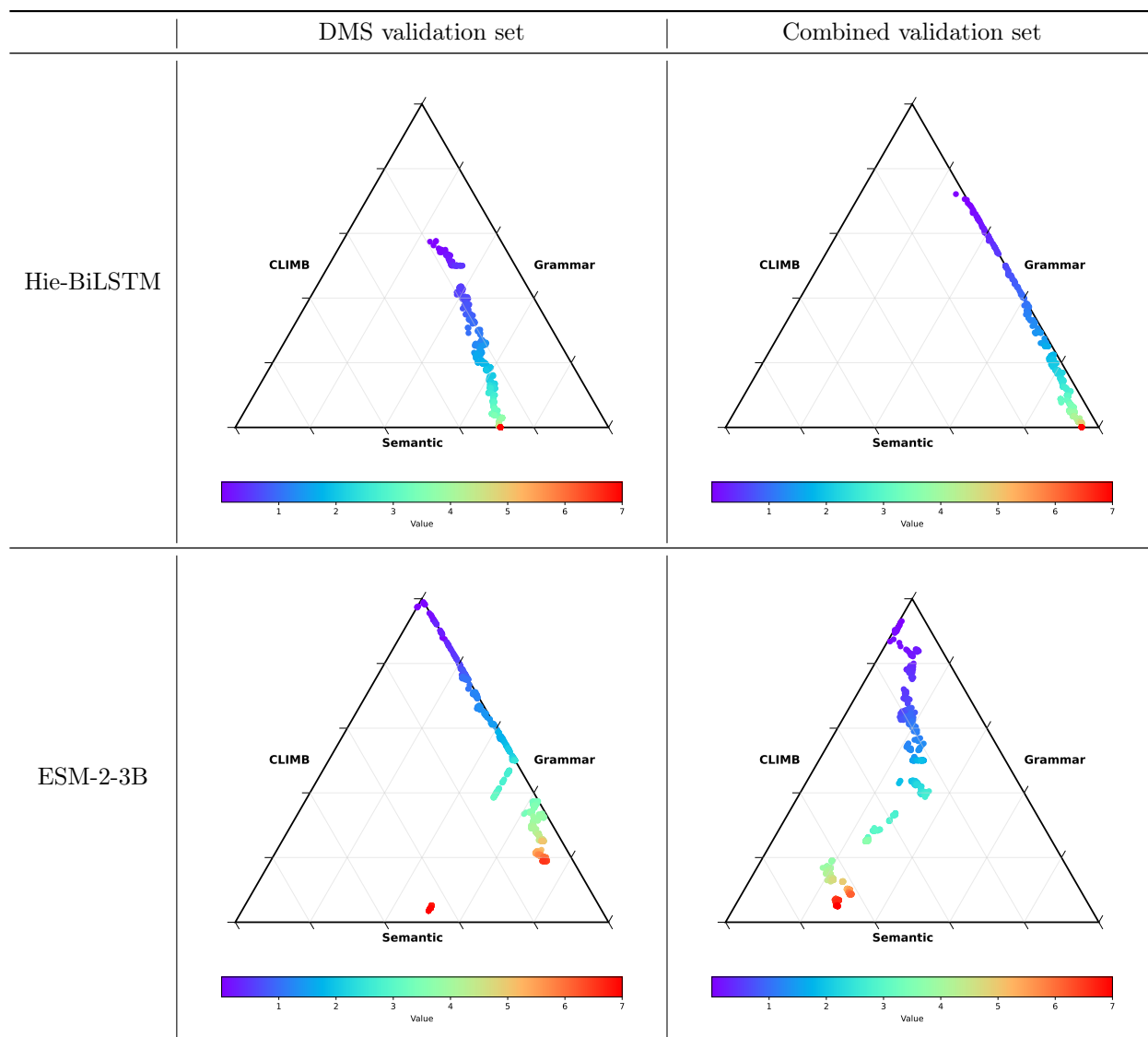


Figure S4: Rank distributions across validation sets. Box plots compare CSCS, CAC, CLIMB-only ablation, and random baselines.

Table S3: Representative optimal-weight trajectories.



S5 Additional landscape analysis

The PRL manuscript emphasizes the model-process mismatch and the horizon-dependent crossover. The domain-level Spike landscape analysis from the long-form manuscript is retained here as diagnostic information. These summaries should be interpreted as model-behavior diagnostics rather than independent biological claims.

Table S4: Domain-resolved average rank shifts for all possible mutations. Negative values indicate promotion by CAC relative to CSCS.

Random						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	11944.52 ± 7004.79	12133.53 ± 6969.43	12123.73 ± 7257.53	12087.71 ± 6901.65	12175.62 ± 7103.16	11885.03 ± 6941.11
Rank _{CAC}	12171.62 ± 7025.52	12080.15 ± 6974.83	12630.38 ± 7010.15	12142.85 ± 6900.83	11685.95 ± 6777.75	12505.88 ± 7120.62
Rank _{CAC} -Rank _{CSCS}	227.10 ± 132.97	-53.37 ± 151.48	506.65 ± 531.07	55.14 ± 262.06	-489.66 ± 315.40	620.85 ± 465.67

Hie						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	10466.20 ± 6778.14	11202.43 ± 6765.50	10149.65 ± 6614.45	14189.60 ± 6694.64	15227.74 ± 6127.75	16996.08 ± 6290.81
Rank _{CAC}	11348.84 ± 6873.78	11390.06 ± 6772.49	12108.50 ± 6787.24	13337.70 ± 7197.38	13155.33 ± 6870.01	15140.41 ± 7737.44
Rank _{CAC} -Rank _{CSCS}	882.64 ± 129.38	187.63 ± 147.07	1958.85 ± 498.80	-851.90 ± 263.93	-2072.41 ± 295.73	-1855.68 ± 466.99

ESM2 150M						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	13023.94 ± 6808.17	11607.07 ± 6531.08	12980.91 ± 7456.75	13490.32 ± 8041.74	9438.89 ± 6976.61	8834.14 ± 5628.47
Rank _{CAC}	12572.51 ± 7145.65	11886.74 ± 6891.00	12471.28 ± 6987.12	12388.43 ± 6968.37	11548.68 ± 6733.08	12837.30 ± 7388.89
Rank _{CAC} -Rank _{CSCS}	-451.43 ± 132.28	279.67 ± 145.86	-509.63 ± 537.83	-1101.90 ± 285.72	2109.79 ± 311.47	4003.16 ± 434.97

ESM2 650M						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	12811.63 ± 6538.17	9723.82 ± 6043.81	13007.99 ± 7165.56	15291.89 ± 7661.06	11530.56 ± 7362.05	8875.89 ± 5540.36
Rank _{CAC}	11855.36 ± 6953.31	11731.33 ± 6895.52	12743.82 ± 7033.22	13017.13 ± 7077.25	12031.24 ± 6870.96	13006.45 ± 7390.47
Rank _{CAC} -Rank _{CSCS}	-956.27 ± 127.92	2007.51 ± 140.87	-264.17 ± 528.45	-2274.75 ± 280.05	500.67 ± 323.50	4130.56 ± 432.54

ESM2 3B						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	10815.43 ± 5820.03	8636.73 ± 5726.55	13979.20 ± 7093.41	17916.36 ± 6658.11	12504.48 ± 7011.04	11767.12 ± 7210.61
Rank _{CAC}	11340.45 ± 6759.74	11234.81 ± 6714.19	13450.11 ± 7191.18	13688.97 ± 7163.69	12675.30 ± 7068.91	13522.60 ± 7594.00
Rank _{CAC} -Rank _{CSCS}	525.02 ± 119.55	2598.08 ± 135.57	-529.09 ± 531.63	-4227.39 ± 262.60	170.82 ± 319.84	1755.48 ± 490.39

S6 Availability and reproducibility

All source code developed for this study is available at <https://github.com/sos440/climb> and archived on Zenodo at <https://doi.org/10.5281/zenodo.15744443>. The same Zenodo archive contains the generated CAC scores, CSCS scores, model component scores, and data needed to reproduce the figures and tables.

The core reproduction workflow is:

1. Clone the repository and install dependencies.

```
$ git clone https://github.com/sos440/climb
$ cd climb
$ conda create --name climb python=3.10 -y
```


Table S5: Domain classification of mutations in the combined validation set.

	Validation sets of escape mutations per domains
NTD	(1) L18F, (2) T19R, (3) A67V, (4) D80A, (5) T95I, (6) G142D, (7) L212I, (8) D215G
RBD	(9) G339D, (10) S371L, (11) S373P, (12) S375F, (13) K417E, (14) K417N, (15) N440K, (16) N440Q, (17) V445A, (18) G446S, (19) N450D, (20) L452R, (21) Y453F, (22) L455F, (23) S477N, (24) T478K, (25) E484A, (26) E484K, (27) G485D, (28) F486V, (29) F490L, (30) F490S, (31) Q493K, (32) Q493R, (33) G496S, (34) Q498R, (35) N501Y, (36) Y505H
FP	(52) D796Y
HR1	(54) D950N, (55) Q954H, (56) N969K, (57) L981F, (58) S982A
HR2	
TM	
Others (Not in domains)	(37) T547K, (38) A670D, (39) D614G, (40) H655Y, (41) N679K, (42) P681H, (43) P681R, (44) R682Q, (45) R685S, (46) V687G, (47) A701V, (48) T716I, (49) N764K, (50) G769E, (51) Q779K, (59) D1118H, (60) V1128A

Table S6: Domain-specific rank changes for the combined validation set. This table focuses only on the 60 known substitutions in the combined set.

Random						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	9871.75 ± 7505.33	12167.21 ± 6381.07	994.00 ± 0.00	14502.40 ± 6923.04	None	None
Rank _{CAC}	5000.38 ± 4667.58	4951.39 ± 3882.43	570.00 ± 0.00	6596.40 ± 6387.73	None	None
Rank _{CAC} -Rank _{CSCS}	-4871.38 ± 3735.63	-7215.82 ± 1411.48	-424.00 ± 0.00	-7906.00 ± 3823.49	None	None

Hie						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	6750.88 ± 7516.32	6165.96 ± 4886.01	1312.00 ± 0.00	9186.20 ± 7027.47	None	None
Rank _{CAC}	4375.88 ± 7437.02	3971.32 ± 3881.44	393.00 ± 0.00	6076.40 ± 5016.69	None	None
Rank _{CAC} -Rank _{CSCS}	-2375.00 ± 3133.41	-2194.64 ± 1158.17	-919.00 ± 0.00	-3109.80 ± 4108.29	None	None

ESM2 150M						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	11047.12 ± 7122.90	7464.82 ± 4899.32	17044.00 ± 0.00	10160.40 ± 8054.21	None	None
Rank _{CAC}	4911.50 ± 4695.86	4774.75 ± 3699.31	585.00 ± 0.00	6564.40 ± 6111.78	None	None
Rank _{CAC} -Rank _{CSCS}	-6135.62 ± 2980.30	-2690.07 ± 1163.19	-16459.00 ± 0.00	-3653.60 ± 4521.60	None	None

ESM2 650M						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	8481.62 ± 6743.16	6187.43 ± 4179.92	21296.00 ± 0.00	9982.80 ± 8609.31	None	None
Rank _{CAC}	4773.88 ± 4507.99	4789.71 ± 3725.69	644.00 ± 0.00	6564.40 ± 6111.78	None	None
Rank _{CAC} -Rank _{CSCS}	-3707.75 ± 2815.26	-1397.71 ± 1032.81	-20652.00 ± 0.00	-3418.40 ± 4667.96	None	None

ESM2 3B						
	NTD	RBD	FP	HR1	HR2	TM
Rank _{CSCS}	7524.50 ± 6152.67	4987.25 ± 3417.16	19945.00 ± 0.00	12459.60 ± 7111.16	None	None
Rank _{CAC}	4453.62 ± 4234.99	4567.21 ± 3520.72	1901.00 ± 0.00	6652.80 ± 5901.62	None	None
Rank _{CAC} -Rank _{CSCS}	-3070.88 ± 2544.85	-420.04 ± 698.70	-18044.00 ± 0.00	-5806.80 ± 3610.76	None	None

```
$ conda activate climb
$ pip install -r requirements.txt
```

2. Generate primary scores.

```
$ cd viral-analyzer
$ python main.py \
  --escape "SARS-CoV-2-WildType>*" \
  --result "SARS-CoV-2-WildType-*" \
  --scores --seed 42
```

3. Optimize component weights.

```
$ python main.py \
  --escape "SARS-CoV-2-WildType>*" \
  --result "SARS-CoV-2-WildType-*" \
  --minimizer --seed 42
```

4. Run the CLIMB-only ablation.

```
$ python main.py \
  --escape "SARS-CoV-2-WildType>*" \
  --result "SARS-CoV-2-WildType-Random" \
  --ablation --seed 42
```

References

- [1] James R. Norris. *Markov Chains*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998.
- [2] Brian Hie et al. “Learning the language of viral evolution and escape”. In: *Science* 371.6526 (2021), pp. 284–288.
- [3] Kijong Yi et al. “Mutational spectrum of SARS-CoV-2 during the global pandemic”. In: *Experimental & Molecular Medicine* 53.8 (Aug. 2021), pp. 1229–1237. ISSN: 1226-3613. DOI: 10.1038/s12276-021-00658-z. URL: <http://dx.doi.org/10.1038/s12276-021-00658-z>.
- [4] Zeming Lin et al. “Evolutionary-scale prediction of atomic-level protein structure with a language model”. In: *Science* 379.6637 (2023), pp. 1123–1130.